

BOUNDING THE MAXIMUM LIKELIHOOD ESTIMATOR IN EXPONENTIAL FAMILIES

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Summary: A method for bounding the maximum likelihood estimator (MLE) for a k -dimensional parameter vector in continuous exponential families is established. This is achieved by showing that the ordinate of the log-likelihood function is a linear function of the entropy at the MLE and by using frequency considerations to establish arbitrarily accurate bounding functions for entropy. The bounding functions involve only the normalising function of the density, permitting derivative free computation of the MLE. Examples are taken from gamma and beta distributions.

1. Introduction

In this paper, certain monotonicity properties of ‘frequency moment’-type functions are used to obtain bounds for the maximum likelihood estimator (MLE) in continuous exponential families. Manas (1997) suggests a method of parameter estimation based on earlier ideas of Yule (1938) and Sichel (1982) involving frequency moments. In particular, Sichel (1982) demonstrates asymptotic optimality of these so-called “frequency moment” estimators. The

estimating equation suggested by Manas is

$$\ell(\hat{\theta}) = -nH(\hat{\theta}), \quad (1.1)$$

where $\hat{\theta}$ is the frequency moment estimator of an unknown parameter θ and $\ell(\theta)$ and $H(\theta)$ denote the log-likelihood function and entropy, respectively.

The paper proceeds with some preliminary, known results on parameter estimation in exponential families. It is then shown that MLEs in this class satisfy (1.1) and bounding functions for entropy are established, from which bounds for the MLE are deduced. An important special case yields bounds of unit width. Examples are drawn from the two-parameter gamma and beta distributions, where these ‘unitary’ bounds yield a line segment and a parallelogram, respectively.

2. Preliminaries

Let X denote a continuous random variable with density function of the form

$$f_X(x, \theta) = \frac{1}{N(\theta)} \exp\left\{\sum_{i=1}^k \theta_i t_i(x)\right\}, \quad (2.1)$$

over a range, R , independent of θ , so that X belongs to the k -parameter exponential family of distributions, where $\theta = (\theta_1, \dots, \theta_k) \in \Theta$ denotes a k -dimensional vector of unknown parameters. The functions $t_i(x)$, $i = 1, \dots, k$, each have only the single argument, x , and also possess density functions of the form (2.1). $N(\theta)$ is the so-called “normalising function” defined by $N(\theta) = \int_R \exp\{\sum_{i=1}^k \theta_i t_i(x)\} dx$. These include a wide class of special functions.

Let $\hat{\theta} = (\hat{\theta}_1, \dots, \hat{\theta}_k)$ denote the MLE of θ from a random sample $\tilde{X} = (X_1, \dots, X_n)$ of X . Lehmann (1959, p.52–58) has investigated

properties of $\hat{\theta}$ in this family. In particular, the likelihood equations are

$$\bar{T}_i \equiv \frac{\partial}{\partial \theta_i} \log N(\theta), \quad (2.2)$$

where $\bar{T}_i = \sum_{s=1}^n t_i(X_s)/n$, for $i = 1, \dots, k$. The vector variable, $\bar{T} = (\bar{T}_1, \dots, \bar{T}_k)$ is complete and minimally sufficient for θ , with

$$E(\bar{T}_i) = \frac{\partial}{\partial \theta_i} \log N(\theta) \quad (2.3)$$

and

$$\text{Cov}(\bar{T}_i, \bar{T}_j) = \frac{1}{n} \frac{\partial^2}{\partial \theta_i \partial \theta_j} \log N(\theta), \quad (2.4)$$

for $i, j = 1, \dots, k$.

From (2.1), the log-likelihood function defined by $\ell_{\bar{X}}(\theta) = \log \Pi_{s=1}^n f_X(x_s, \theta)$, becomes

$$\ell_{\bar{X}}(\theta) = n \{ -\log N(\theta) + \sum_{i=1}^k \theta_i \bar{T}_i \}, \quad (2.5)$$

and $\hat{\theta}$ is such that $\max_{\theta} \ell_{\bar{X}}(\theta) = \ell_{\bar{X}}(\hat{\theta})$, obtained at the solutions to the likelihood equations, (2.2).

It is important to note that $\ell_{\bar{X}}(\theta)$ is strictly concave over Θ since

$$\frac{\partial^2}{\partial \theta_i^2} \ell_{\bar{X}}(\theta) = -n \frac{\partial^2}{\partial \theta_i^2} \log N(\theta) = -n^2 \text{Var}(\bar{T}_i) < 0, \quad (2.6)$$

and the $k \times k$ matrix of second derivatives with (i, j) -th entry,

$$\frac{\partial^2}{\partial \theta_i \partial \theta_j} \ell_{\bar{X}}(\theta) = -n \frac{\partial^2}{\partial \theta_i \partial \theta_j} \log N(\theta) = -n^2 \text{Cov}(\bar{T}_i, \bar{T}_j), \quad (2.7)$$

is negative definite by (2.4). It follows that any $\hat{\theta}$ satisfying (2.2) is a unique global maximum of $\ell_{\bar{X}}(\theta)$.

Now, Shannon's entropy, defined by $H_X(\theta) = -E\{\log f_X(X, \theta)\}$, becomes

$$H_X(\theta) = \log N(\theta) - \sum_{i=1}^k \theta_i \frac{\partial}{\partial \theta_i} \log N(\theta), \quad (2.8)$$

from (2.3). This is a well-known measure of diversity, commonly used in information theory (see e.g. Kotz, 1966, p.3ff). It follows that the maximum likelihood estimator must satisfy (1.1), which is the estimator derived from frequency moment considerations in Manas (1997).

It is of interest to note that the estimator defined in (1.1) minimises ‘observed’ entropy, defined by

$$\tilde{H}_X(\theta) = \log N(\theta) - \sum_{i=1}^k \theta_i \bar{T}_i, \quad (2.9)$$

i.e. $\min_{\theta} \tilde{H}_X(\theta) = \tilde{H}_X(\hat{\theta})$.

3. Bounds

Define the functions

$$\omega_X(s, \theta) = \int_{-\infty}^{\infty} \{f_X(x, \theta)\}^{s+1} dx \quad (3.1)$$

and

$$\pi_X(s, \theta) = \frac{1}{s} \log \omega_X(s, \theta). \quad (3.2)$$

(3.1) defines the frequency moment of X of order s and has been studied by Moriguti (1952) and Golomb (1966). The former establishes the infimum for ω_X over X and the latter treats it as a generating function, the so-called “information generating function”. It has also been extensively used in parameter estimation (e.g. Yule, 1938, Sichel, 1947, 1949, 1967, 1982). The functions (3.1) and (3.2) are typically defined over a set of real s -values of the form $(-a, \infty)$, where $a > 0$. Clearly $\omega_X(s, \theta) > 0$ and $\omega_X(0, \theta) = 1$.

The bounds themselves are derived from the following two lemmas, which are proved in the appendix (Section 6).

Lemma 1.

$$\lim_{s \rightarrow 0} \pi_X(s, \theta) = -H_X(\theta)$$

Lemma 2. For given θ , $\pi_X(s, \theta)$ is monotone increasing in s .

These two lemmas provide the following result:

Theorem. For given θ and any $s^+ > 0$ and $s^- < 0$,

$$\pi_X(s^-, \theta) \leq -H_X(\theta) \leq \pi_X(s^+, \theta). \quad (3.3)$$

In addition, for any $\epsilon > 0$, there exists some $\delta > 0$ such that $|\pi_X(s, \theta) + H_X(\theta)| < \epsilon$ whenever $|s| < \delta$.

Now, let $\hat{\theta} \cdot \bar{T} = \sum_{i=1}^k \hat{\theta}_i \bar{T}_i$ denote the scalar product of the MLE, $\hat{\theta}$ with the sufficient statistics, $\bar{T}$.

Now, from (3.2) and (2.1) it follows that

$$\pi_X(s, \theta) = b_X(s, \theta) - \log N(\theta), \quad (3.4)$$

where

$$b_X(s, \theta) = \frac{\log N((s+1)\theta) - \log N(\theta)}{s} \quad (3.5)$$

is increasing in s by lemma 2. In addition, lemma 1 gives

$$\lim_{s \rightarrow 0} b_X(s, \theta) = \sum_{i=1}^k \theta_i \frac{\partial}{\partial \theta_i} \log N(\theta), \quad (3.6)$$

a known result in the theory of directional derivatives (see e.g. Edwards, 1973, p. 69).

The bounds in theorem (3.3) thus become

$$b_X(s^-, \hat{\theta}) \leq \hat{\theta} \cdot \bar{T} \leq b_X(s^+, \hat{\theta}), \quad (3.7)$$

where $-a < s^- < 0 < s^+$ with equality in the limit as $s^-, s^+ \rightarrow 0$, and in particular

$$b_X(-s, \hat{\theta}) \leq \hat{\theta} \cdot \bar{T} \leq b_X(s, \hat{\theta}) \quad (3.8)$$

for any $s > 0$.

Next, we derive bounds for $\hat{\theta}$ itself. We initially assume that θ is one dimensional (i.e. $k = 1$) and (w.l.o.g.) we also assume that $\hat{\theta} > 0$. (3.8) then reduces to

$$\frac{\log N(\hat{\theta} - \hat{\theta}s) - \log N(\hat{\theta})}{-\hat{\theta}s} \leq \bar{T} \leq \frac{\log N(\hat{\theta} + \hat{\theta}s) - \log N(\hat{\theta})}{\hat{\theta}s}. \quad (3.9)$$

Letting $s = 1/\hat{\theta}$, in (3.9) gives

$$\log A(\hat{\theta}) < \bar{T} < \log A(\hat{\theta} + 1), \quad (3.10)$$

where

$$A(\theta) = N(\theta)/N(\theta - 1). \quad (3.11)$$

Since (3.10) applies for any $\hat{\theta}$, it follows that A is monotone increasing over Θ and hence

$$A^{-1}(e^{\bar{T}}) - 1 < \hat{\theta} < A^{-1}(e^{\bar{T}}), \quad (3.12)$$

restricting $\hat{\theta}$ to an interval of unit width.

For $k > 1$, let $\theta_i^* = (\theta_1, \dots, \theta_{i-1}, \theta_{i+1}, \dots, \theta_k)$ and for each θ_i , ($i = 1, \dots, k$), suppose, initially, that θ_i^* is known. It follows from (3.11) and (3.12) that the maximum likelihood estimator, $\hat{\theta}_i$ must lie in the interval, Θ_i^* , say, defined by

$$\Theta_i^* = \{\theta_i : A_i^{-1}(e^{\bar{T}_i}) - 1 < \theta_i < A_i^{-1}(e^{\bar{T}_i})\}, \quad (3.13)$$

for $i = 1, \dots, k$, where

$$A_i(\theta_i) = N(\theta)/N(\theta_1, \dots, \theta_{i-1}, \theta_i - 1, \theta_{i+1}, \dots, \theta_k). \quad (3.14)$$

The interval Θ_i^* restricts the range of $\hat{\theta}_i$ in terms of θ_i^* . It follows that

$$\hat{\theta} \in \Theta^* = \bigcap_{i=1}^k \Theta_i^*. \quad (3.15)$$

Clearly $A_i^{-1}(e^{\bar{T}_i})$ is a function of $\bar{T}_i$ and θ_i^* and (3.15) defines a parallelopiped in $\Re^k$ of unit width in the direction of any axis.

To obtain a ‘geometric’ picture of the bounds for $\hat{\theta}$ derived from (3.8), let $s > 0$ be given and define the following sets of ordinates indexed by θ :

$$B_\theta(s) = \{b : b_X(-s, \theta) \leq b \leq b_X(s, \theta)\}, \quad (3.16)$$

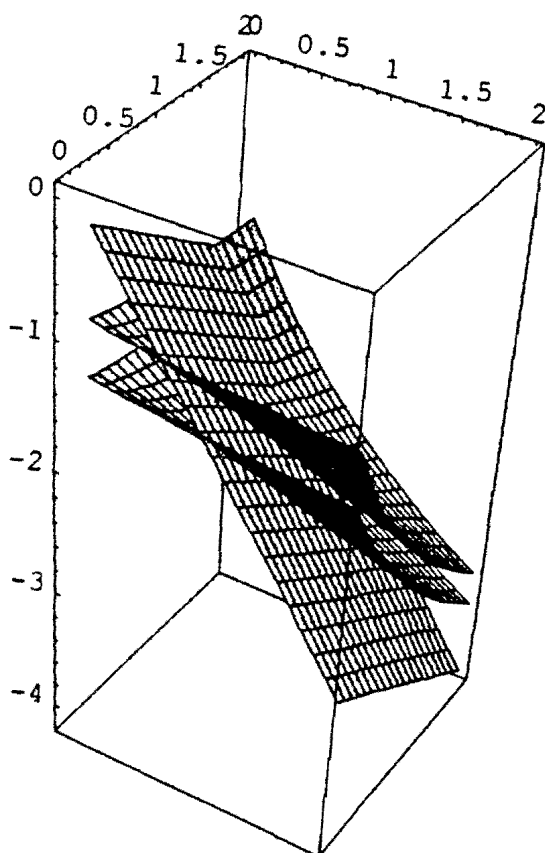
and

$$C_\theta = \{c(\theta) : c(\theta) = \theta \cdot \bar{T}\}. \quad (3.17)$$

$B_\theta(s)$ is a ‘band’ or lamina of width, $b_X(s, \theta) - b_X(-s, \theta)$, which can be made arbitrarily small by choosing s sufficiently close to 0. In fact, as $s \rightarrow 0$, $B_\theta(s)$ approaches the function (3.6), which is independent of the sample. C_θ is a plane through the origin, which depends on the sample through the sufficient statistics. It follows that $\hat{\theta}$ lies in the inverse image of $B_\theta(s) \cap C_\theta$, i.e.

$$\hat{\theta} \in \{\theta : c(\theta) \in B_\theta(s) \cap C_\theta\}. \quad (3.18)$$

Figures 1, 2 and 3 illustrate (3.16) and (3.17) for successively decreasing values of s for a simulated sample from the beta distribution, for which $k = 2$.

**Figure 1**

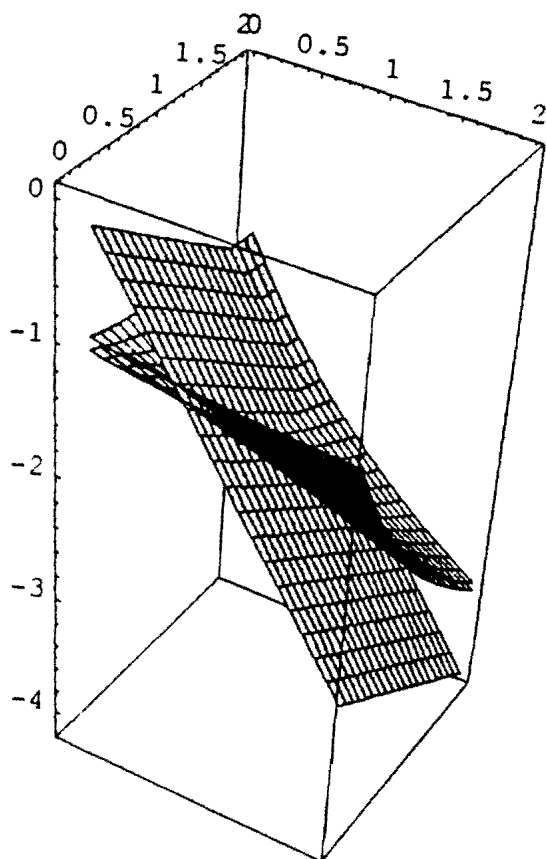
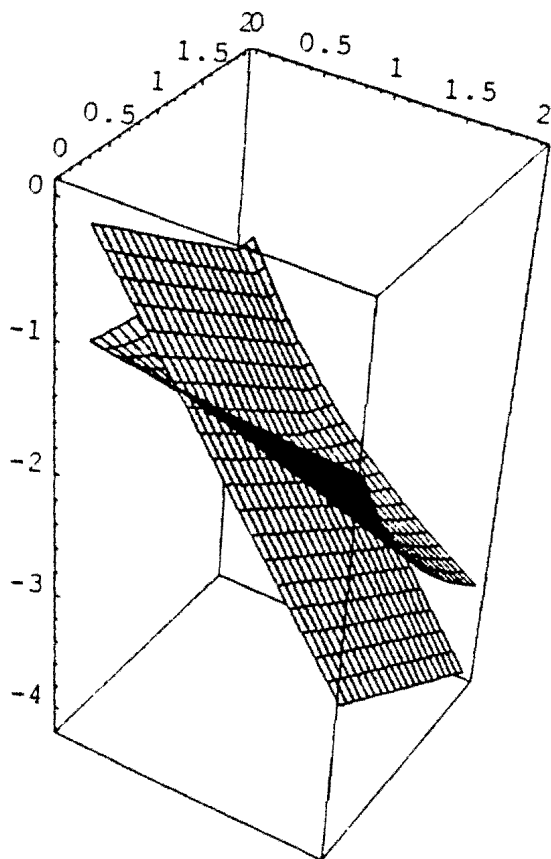


Figure 2

**Figure 3**

4. Examples

1. GAMMA DISTRIBUTION

Consider the gamma density function, which, for convenience will be parameterised thus:

$$f_X(x, \theta_1, \theta_2) = \frac{\theta_2^{\theta_1+1}}{\Gamma(\theta_1+1)} x^{\theta_1} e^{-\theta_2 x}, \quad (4.1)$$

where $x \geq 0$, $\theta_1 > -1$ and $\theta_2 > 0$. The normalising function is $N(\theta) = \Gamma(\theta_1+1)/\theta_2^{\theta_1+1}$. Also, $t_1(x) = \log x$ and $t_2(x) = -x$, so that sufficient statistics for (θ_1, θ_2) are $(\log \tilde{x}, \bar{x})$, where $\tilde{x} = \{\Pi_{s=1}^n x_s\}^{1/n}$ and $\bar{x} = \sum_{s=1}^n x_s/n$, the sample geometric and arithmetic means, respectively.

The entropy of X is

$$H_X(\theta) = -\log \frac{\theta_2}{\Gamma(\theta_1+1)} - \theta_1 \psi(\theta_1+1) + \theta_1 + 1, \quad (4.2)$$

where ψ is the digamma (psi) function defined by $\psi = \Gamma'/\Gamma$. From theorem (3.3), it follows that $-H_X(\theta)$ is bounded by the function

$$\pi_X(s, \theta) = \log \frac{\theta_2}{\Gamma(\theta_1+1)} + h(s, \theta), \quad (4.3)$$

where

$$h(s, \theta) = \frac{1}{s} \left[\log \frac{\Gamma(\theta_1(s+1)+1)}{\Gamma(\theta_1+1)} - \{\theta_1(s+1)+1\} \log(1+s) \right]. \quad (4.4)$$

The likelihood equations are

$$\log \tilde{x} = \psi(\hat{\theta}_1+1) - \log \hat{\theta}_2 \quad (4.5)$$

and

$$\bar{x} = (\hat{\theta}_1+1)/\hat{\theta}_2, \quad (4.6)$$

which satisfy (1.1). In addition, from (3.8),

$$b_X(-s, \hat{\theta}) < \hat{\theta}_1 \log \tilde{x} - \hat{\theta}_2 \bar{x} < b_X(s, \hat{\theta}) \quad (4.7)$$

for any $s > 0$, where $b_X(s, \theta) = -(\theta_1+1) + h(s, \theta)$, following (3.5).

We need only consider bounds for $\hat{\theta}_1$ since it specifies $\hat{\theta}_2$ from (4.6). Following (3.14), $A(\theta_1) = N(\theta_1, \theta_2)/N(\theta_1 - 1, \theta_2) = \theta_1/\theta_2$, since $\Gamma(\theta_1 + 1) = \theta_1\Gamma(\theta_1)$. Thus $A^{-1}(u) = \theta_2 u$, and hence from (3.12), $\hat{\theta}_2 \tilde{x} - 1 < \hat{\theta}_1 < \hat{\theta}_2 \tilde{x}$, from which it follows from (4.6) that

$$-1 < \hat{\theta}_1 < \frac{\tilde{x}}{\bar{x} - \tilde{x}}. \quad (4.8)$$

The midpoint of this interval provides an excellent initial estimate and indicates that large values of $\hat{\theta}_1$ are associated with small values of $\bar{x} - \tilde{x}$ and vice versa.

Perhaps a more familiar form of this density is

$$f_X(x, r, \lambda) = \frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}, \quad (4.9)$$

for $x \geq 0$ and $r, \lambda > 0$, for which $\theta_1 = r - 1$ and $\theta_2 = \lambda$ in the preceding analysis. Application to the normalising function, $N(\lambda, r) = \Gamma(r)/\lambda^r$ yields obvious variations on the above results and in particular, for given λ , $A(r) = (r - 1)/\lambda$, so that $A^{-1}(u) = \lambda u + 1$ and (3.12) reduces to $\hat{\lambda} \tilde{x} < \hat{r} < \hat{\lambda} \tilde{x} + 1$ so that $(\hat{\lambda}, \hat{r})$ lies on the line segment between the origin and $(1, \bar{x})/(\bar{x} - \tilde{x})$.

For this parameterisation,

$$\begin{aligned} b_X(s, \lambda, r) &= \frac{1}{s} \{ \log \Gamma(r(1+s)) / \{ \lambda(1+s) \}^{r(1+s)} - \log \Gamma(r) / \lambda^r \} \\ &= \frac{1}{s} \log \frac{\Gamma(r(1+s))}{\Gamma(r)} - r \log \lambda - r(1 + \frac{1}{s}) \log(1+s), \end{aligned} \quad (4.10)$$

and hence $\lim_{s \rightarrow 0} b_X(s, r, \lambda) = r(\psi(r) - \log \lambda - 1)$. The likelihood equations are $\log \tilde{x} = \psi(\hat{r}) - \log \hat{\lambda}$ and $\bar{x} = \hat{r}/\hat{\lambda}$, from which it follows that

$$b_X(-s, \hat{r}, \hat{\lambda}) < \hat{r} \log \tilde{x} - \hat{\lambda} \bar{x} < b_X(s, \hat{r}, \hat{\lambda}), \quad (4.11)$$

for any $s > 0$, with $\hat{\lambda} \bar{x} = \hat{r}$.

For example, $\tilde{x} = 4.5091$ and $\bar{x} = 5.0000$, were observed from a simulated sample. The MLE, $(\hat{\lambda}, \hat{r})$ lies on the line segment between $(0, 0)$ and $(2.0371, 10.1853)$, the midpoint of which provides the initial estimate $(1.0186, 5.0925)$, which is remarkably close to the MLE. In this case, the MLE is the (unique) solution to (3.8) in the limit as $s \rightarrow 0$. In particular, Table 1 provides values of the bounds on $\hat{r}$ obtained from (4.11) for diminishing s . From this we find (correct to 4 decimals) that $\hat{r} = 4.9991$ and $\hat{\lambda} = 0.9998$.

s	Interval containing $\hat{r}$
0.1	(4.7647, 5.2672)
0.01	(4.9742, 5.0242)
0.001	(4.9966, 5.0016)
0.0001	(4.9988, 4.9993)
0.00001	(4.9991, 4.9992)
0.000005	(4.9991, 4.9991)

2. BETA DISTRIBUTION

Suppose that

$$f_X(x, \theta) = \frac{1}{B(\theta_1 + 1, \theta_2 + 1)} x^{\theta_1} (1 - x)^{\theta_2}, \quad (4.12)$$

where $0 \leq x \leq 1$ and $\theta_1, \theta_2 > -1$. The statistics $(\tilde{x}, \tilde{x}')$ are jointly sufficient for θ , where $\tilde{x}$ is the sample geometric mean and $\tilde{x}' = \{\prod_{j=1}^n (1 - x_j)\}^{1/n}$. The normalising function, $N(\theta) = B(\theta_1 + 1, \theta_2 + 1) = \Gamma(\theta_1 + 1)\Gamma(\theta_2 + 1)/\Gamma(\theta_1 + \theta_2 + 2)$, so that for θ_2 given, $A_1(\theta_1) = B(\theta_1 + 1, \theta_2 + 1)/B(\theta_1, \theta_2 + 1) = \theta_1/(\theta_1 + \theta_2 + 1)$ and thus $A_1^{-1}(u) = (\theta_2 + 1)u/(1 - u)$. Hence from (3.13)

$$(\theta_2 + 1)\bar{h} - 1 < \hat{\theta}_1 < (\theta_2 + 1)\tilde{h}, \quad (4.13)$$

where $\tilde{h} = \tilde{x}/(1 - \tilde{x})$ and similarly, for θ_1 known,

$$(\theta_1 + 1)\tilde{h}' - 1 < \hat{\theta}_2 < (\theta_1 + 1)\tilde{h}', \quad (4.14)$$

where $\tilde{h}' = \tilde{x}'/(1 - \tilde{x}')$.

Thus the set, Θ^* in (3.15), is a parallelogram in the (θ_1, θ_2) -plane with bottom left vertex at $(-1, -1)$ and extending diagonally upward along the direction of maximum ascent of the likelihood function to the point $(\tilde{h}(1 + \tilde{h}'), \tilde{h}'(1 + \tilde{h}))/ (1 - \tilde{h}\tilde{h}')$. The centre of this parallelogram provides an excellent initial estimate in a Newton-Raphson or 'bisection' algorithm.

The plane, $c(\theta)$ defined in (3.17) is $c(\theta) = \theta_1 \log \tilde{x} + \theta_2 \log \tilde{x}'$ and the bounds, (3.8) become $b_X(-s, \hat{\theta}) \leq c(\hat{\theta}) \leq b_X(s, \hat{\theta})$, where

$$b_X(s, \theta) = \frac{1}{s} \log \frac{B(\theta_1(s+1) + 1, \theta_2(s+1) + 1)}{B(\theta_1 + 1, \theta_2 + 1)}, \quad (4.15)$$

for $s > 0$. These bounds constitute a tight check on whether the likelihood has indeed been maximised at $\hat{\theta}$, particularly for small s .

For example suppose that a sample of observations, 0.1; 0.1; 0.2; 0.2; 0.2; 0.4; 0.6; 0.8, is observed from a beta distribution. The sufficient statistics are then $\tilde{x} = 0.2502$ and $\tilde{x}' = 0.6129$. The parallelogram defined by (4.13) and (4.14) extends from $(-1, -1)$ to $(1.8165, 4.4771)$ with the other vertices at $(-0.30, 1.12)$ and $(1.12, 2.36)$. Maximising the log-likelihood over this parallelogram quickly yields the maximum likelihood estimates $\hat{\theta}_1 = 0.2707$ and $\hat{\theta}_2 = 1.4793$.

For $s = 0.0001$, we get from (4.15) that $b_X(-s, \hat{\theta}) = -1.09930$, $b_X(s, \hat{\theta}) = -1.09928$ and $c(\hat{\theta}) = -1.09929$, verifying maximisation of the likelihood. (Several more decimals of accuracy were used in the computations than those quoted. It seems that by letting $s = 10^{-d}$, (4.15) can verify $\hat{\theta}$, accurate to d decimal places, constrained by machine accuracy, which breaks down as s approaches 0.)

As in the gamma example, the reparameterisation, $a = \theta_1 + 1$ and $b = \theta_2 + 1$, suggests examining the normalising function in the form $B(a, b)$,

again providing variants on the above results, and in particular a parallelogram, Θ^* which is shifted one unit in the direction of maximum ascent, to 'begin' at the origin.

5. Concluding remarks

In exponential families, estimators satisfying (1.1) coincide with MLE and this feature is used to establish the bounds from frequency moment based bounding functions for entropy. These are useful in establishing bounded search regions containing the MLE as well as accurate initial estimates, particularly useful in iterative routines which are dependent on 'good' starting values. Several algorithms are available for maximisation of this kind, e.g. Nelder and Mead (1965). In addition, the theory also provides arbitrarily accurate bounding functions, based only on the normalising function, for a linear combination of the estimator and the sufficient statistics, establishing a check that $\ell(\theta)$ is indeed maximised at $\hat{\theta}$.

Although this approach was illustrated only in the case of the gamma and beta distributions, it clearly suggests itself to more complex members of the exponential family, such as the generalised inverse Gaussian (GIG) and its logarithmic transform, the log-GIG distribution (Jørgensen, 1982). The latter was of particular interest to the late H.S. Sichel on account of its great flexibility and hence applicability in data modelling. Boyd (1997) gives exact parameterisations (or limiting cases thereof) which provide a variety of different shapes, from symmetric to highly skew and from leptokurtic to platykurtic.

The log-GIG distribution was used to good effect to model geological data on mining deposits (Sichel, Kleingeld and Assibey-Bonsu, 1992; Dohm, 1995 and Sichel, Dohm and Kleingeld, 1997). In these papers, however, method-

of-moments estimation was used, presumably due to computational difficulties involved in differentiating or maximising the Bessel- K function with respect to its order. Clearly MLEs would be preferable here to their large sample optimality properties, particularly in view of the large sample sizes reported by these authors.

Preliminary investigations into this member of the two parameter exponential family look promising and should simplify the computation of MLEs, improving the fits obtained by these authors. The results of this research will be reported in a sequel.

6. Appendix

PROOF OF LEMMA 1

$\pi_X(s, \theta)$ is clearly indeterminate as $s \rightarrow 0$. Thus from L'Hospital's rule,

$$\lim_{s \rightarrow 0} \pi_X(s, \theta) = \lim_{s \rightarrow 0} \int_{-\infty}^{\infty} \{f_X(x, \theta)\}^{s+1} \log f_X(x, \theta) dx / \omega_X(s, \theta) = -H_X(\theta).$$

PROOF OF LEMMA 2

We apply Jensen's well-known inequality which states that if h and ϕ are real-valued functions defined on $(-\infty, \infty)$ where h is convex (i.e. $h'' \geq 0$) then

$$h\left(\int_{-\infty}^{\infty} \phi(x) f_X(x, \theta) dx\right) \leq \int_{-\infty}^{\infty} h(\phi(x)) f_X(x, \theta) dx.$$

Letting $t \leq s$ in the case $s > 0$ with $h(u) = u^{s/t}$ gives

$$\omega_X(s, \theta) = \int_{-\infty}^{\infty} \{f_X^t(x, \theta)\}^{s/t} f_X(x, \theta) dx \geq \left\{ \int_{-\infty}^{\infty} f_X^{t+1}(x, \theta) dx \right\}^{s/t}.$$

Extracting the s -th root and taking logarithms gives $\pi_X(s, \theta) \geq \pi_X(t, \theta)$. The case $s < 0$ is proven similarly with appropriate sign changes and $h(u) = u^{t/s}$.

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